HDOC DAY 66

Recursive Multiplication of Square Matrices

Aim Multiply two n x n matrices using recursive multiplication logic, without iterative loops in the multiplication function.

Algorithm

1. Read n and the two square matrices.

2. Initialize the result matrix to zero.

3. Use recursive function multiply() with row, column and inner-product index k.

4. Accumulate A[row][k] * B[k][col]; advance k recursively.

5. When k reaches n, advance to the next column; then to the next row.

6. Display the product matrix.

C Program

```
#include <stdio.h>

void multiply(int n, int a[n][n], int b[n][n],
              long long c[n][n], int row, int col, int k)
{
    if (row == n) return;
    if (col == n) {
        multiply(n, a, b, c, row + 1, 0, 0);
        return;
    }
    if (k == n) {
        multiply(n, a, b, c, row, col + 1, 0);
        return;
    }
    c[row][col] += (long long)a[row][k] * b[k][col];
    multiply(n, a, b, c, row, col, k + 1);
}

int main(void)
{
    int n;
    printf("Enter matrix order: ");
    if (scanf("%d", &n) != 1 || n < 1 || n > 20) {
        printf("Invalid order.\n"); return 1;
    }
    int a[n][n], b[n][n];
    long long c[n][n];
    printf("Enter matrix A elements row-wise:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    printf("Enter matrix B elements row-wise:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++) {
            scanf("%d", &b[i][j]);
            c[i][j] = 0;
        }
    multiply(n, a, b, c, 0, 0, 0);
    printf("Product matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++)
            printf("%lld ", c[i][j]);
        printf("\n");
    }
    return 0;
```

```
}
```

Test Cases

Input	Expected result
n=1; A=[3], B=[4]	[12]
n=2; A=[1 2;3 4], B=[5 6;7 8]	[19 22;43 50]
n=2; A=identity, B=[2 3;4 5]	[2 3;4 5]

Compilation and Execution (Mac Terminal)

```
clang -std=c99 -Wall -Wextra day66.c -o day66
./day66
```

Sample Output

```
Enter matrix order: 2
Enter matrix A elements row-wise:
1 2 3 4
Enter matrix B elements row-wise:
5 6 7 8
Product matrix:
19 22
43 50
```

Important Note

Loops are used only for input, initialization and display; all multiplication calculations are recursive.

Result

The program was implemented using the required technique and produces the expected results for the given test cases.